Mini-Project 1: Image Classification with Caltech-101 Dataset

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1 Introduction

This report presents a systematic comparative study of deep learning and classical machine learning approaches for multi-class image classification on the Caltech-101 benchmark dataset, which contains 101 object categories and approximately 9,000 images. Three methods were implemented and rigorously evaluated: ResNet-18 and EfficientNet-B0 (both fine-tuned from ImageNet-pretrained weights), and a Support Vector Machine (SVM) evaluated with two feature extraction strategies (HOG descriptors and frozen ResNet-18 CNN embeddings). All source code used in this project are publicly available at `https://github.com/cathyliu0510/SHBT261`.

2 Method

2.1 Dataset and Preprocessing

The Caltech-101 dataset contains images from 101 object categories plus a `BACKGROUND_Google` distractor class, which was excluded because it does not represent a valid object category. As shown in Figure 1, the dataset exhibits substantial class imbalance, with class counts ranging from roughly 40 to over 800 images. This motivates the use of class-weighted loss and macro-averaged metrics to avoid majority-class dominance.

I applied a stratified 70/15/15 split to preserve class proportions across training, validation,

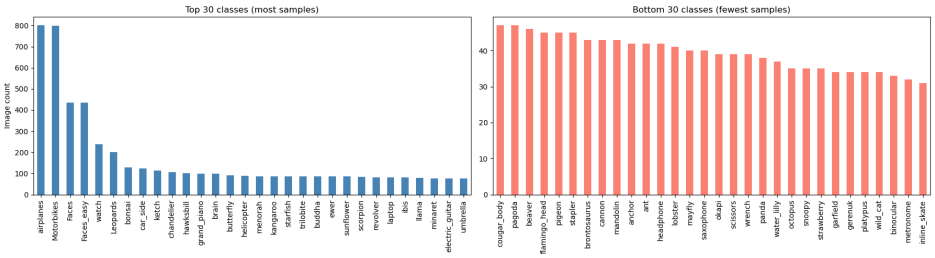


Figure 1: Class distribution in Caltech-101.

and test sets, resulting in 6,073 training images, 1,302 validation images, and 1,302 test images. To mitigate class imbalance during training, I used class-weighted cross-entropy loss, where weights were computed using training labels only via `sklearn`'s `compute_class_weight` with the `balanced` strategy. All images were normalized using ImageNet statistics (mean $= [0.485, 0.456, 0.406]$, std $= [0.229, 0.224, 0.225]$). For deep learning models, the training pipeline optionally applied data augmentation consisting of RandomHorizontalFlip, RandomRotation ($\pm 15°$), ColorJitter, and RandomResizedCrop.

2.2 Model

ResNet-18. ResNet-18 is a residual convolutional neural network that uses skip connections to mitigate vanishing gradient issues in deep architectures. I initialized the network with ImageNet-pretrained weights and replaced the final fully connected layer with Dropout(0.4) followed by Linear$(512, 101)$ to match the target classes. The entire network was fine-tuned end-to-end. Training used the Adam optimizer (learning rate $= 10^{-4}$, weight decay $= 10^{-4}$) with a CosineAnnealingLR schedule over 50 epochs and batch size 32. Class-weighted CrossEntropyLoss was applied to address label imbalance.

EfficientNet-B0. EfficientNet-B0 employs compound scaling to jointly balance network depth, width, and resolution, yielding a parameter-efficient architecture. Its MBConv blocks with squeeze-and-excitation modules enable channel-wise feature recalibration. I initialized EfficientNet-B0 with ImageNet-pretrained weights and replaced the classifier head with Dropout(0.4) followed by Linear$(1280, 101)$, reflecting its larger penultimate feature dimension relative to ResNet-18. To ensure a fair architectural comparison, the optimizer, learning rate schedule, batch size, and loss function were kept identical to the ResNet-18 setup.

Classical ML: SVM with HOG and CNN Features. For the classical pipeline, I trained a Support Vector Machine with RBF kernel ($C = 10$, $\gamma =$ "scale") implemented via a scikit-learn Pipeline.

SVM_HOG. Images were resized to 128×128 and processed using Histogram of Oriented

Gradients (HOG) with 9 orientation bins, 8×8 pixel cells, and 2×2 cell blocks, producing 8,100-dimensional feature vectors. PCA reduced the representation to 512 components prior to SVM training to improve computational efficiency and numerical conditioning.

SVM_CNN. Images were passed through a frozen ImageNet-pretrained ResNet-18 (final fully connected layer removed) to produce 512-dimensional embeddings. PCA was not applied because the learned representation is already compact and well-conditioned.

2.3 Ablation Design

To systematically evaluate model behavior, I conducted controlled ablation studies along three dimensions: input resolution, data augmentation, and feature representation.

Input resolution. For deep learning models, images were resized to 64×64, 128×128, and 224×224 to assess sensitivity to spatial detail. All other training settings were held fixed when varying resolution.

Data augmentation. For ResNet-18 and EfficientNet-B0, I compared training with and without the augmentation pipeline (RandomHorizontalFlip, RandomRotation, ColorJitter, RandomResizedCrop) at the 224×224 resolution. This isolates the regularization effect of augmentation under full fine-tuning.

Feature representation (classical ML pipeline). For the SVM baseline, I compared two feature extractors: hand-crafted HOG descriptors and frozen CNN embeddings from ImageNet-pretrained ResNet-18. The classifier and its hyperparameters were kept identical to isolate the impact of representation quality.

Across all ablations, the optimizer, learning rate schedule, batch size, and training duration were kept constant to ensure fair comparison.

2.4 Evaluation Metrics

All methods were evaluated on the held-out test set using overall accuracy and Top-5 accuracy. To assess performance under class imbalance, I report macro-averaged and weighted precision, re-

call, and F1 scores. Model error patterns are further analyzed using normalized confusion matrices and per-class performance diagnostics.

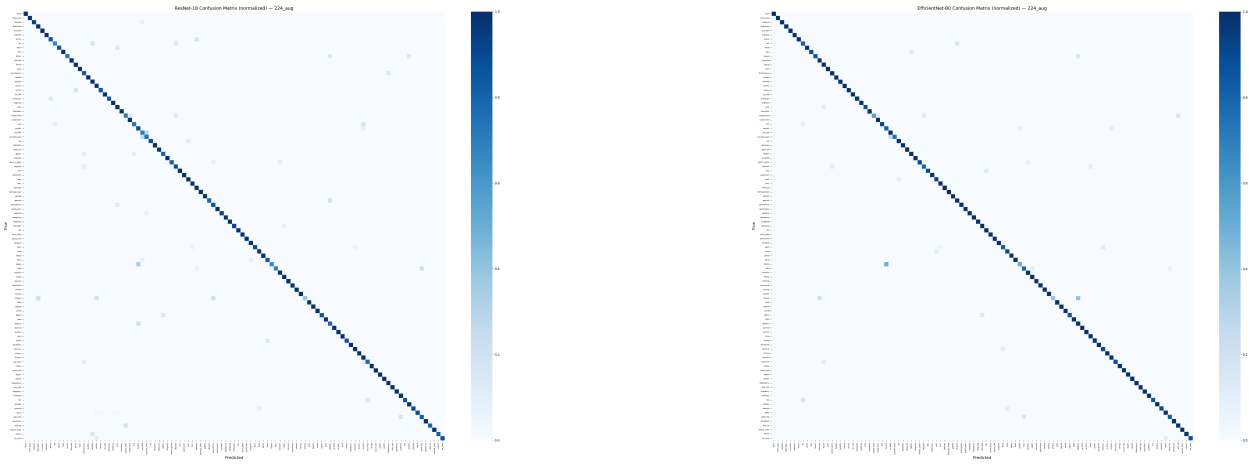
3 Results

ResNet-18

Table 1: ResNet-18 ablation results on the test set.

Config	Accuracy	Top-5 Acc	Macro F1	Weighted F1
224_aug	0.9555	0.9962	0.9317	0.9555
128_aug	0.9409	0.9900	0.9080	0.9389
64_aug	0.8587	0.9578	0.7857	0.8542
224_no_aug	**0.9685**	**0.9962**	**0.9482**	**0.9673**

Table 1 reports the test performance of ResNet-18 under different resolution and augmentation settings. The best configuration (224_no_aug) achieves 96.85% accuracy and 99.62% Top-5 accuracy, with strong Macro and Weighted F1 scores, indicating balanced performance across classes. Precision and recall trends closely mirror those observed in the F1 scores.



(a) ResNet-18 at 224×224

(b) EfficientNet-B0 at 224×224

Figure 2: Normalized confusion matrices for the deep learning models under the 224×224.

The confusion matrix in Figure 2a shows clear diagonal dominance, with limited off-diagonal mass concentrated among visually similar categories.

Ablation summary. Performance remains stable between 224×224 and 128×128 but drops substantially at 64×64, confirming sensitivity to spatial detail. At 224×224, the non-augmented configuration outperforms the augmented variant by approximately 1.3 percentage points in accuracy, indicating that heavy augmentation provides limited additional benefit under full fine-tuning for this architecture.

EfficientNet-B0

Table 2: EfficientNet-B0 ablation results on the test set.

Config	Accuracy	Top-5 Acc	Macro F1	Weighted F1
224_aug	**0.9724**	**0.9977**	**0.9543**	**0.9716**
128_aug	0.9562	0.9962	0.9323	0.9554
64_aug	0.8571	0.9624	0.7893	0.8556
224_no_aug	0.9724	0.9977	0.9522	0.9714

Table 2 reports the test performance of EfficientNet-B0 across ablation settings. The best top-line accuracy (97.24%) and Top-5 accuracy (99.77%) are achieved at 224×224, with identical results for both augmented and non-augmented configurations. Macro and Weighted F1 scores are also high, indicating balanced class performance. Precision and recall trends closely mirror those observed in the F1 scores.

The confusion matrix in Figure 2b shows strong diagonal concentration with minimal off-diagonal mass, indicating slightly improved class separability relative to ResNet-18.

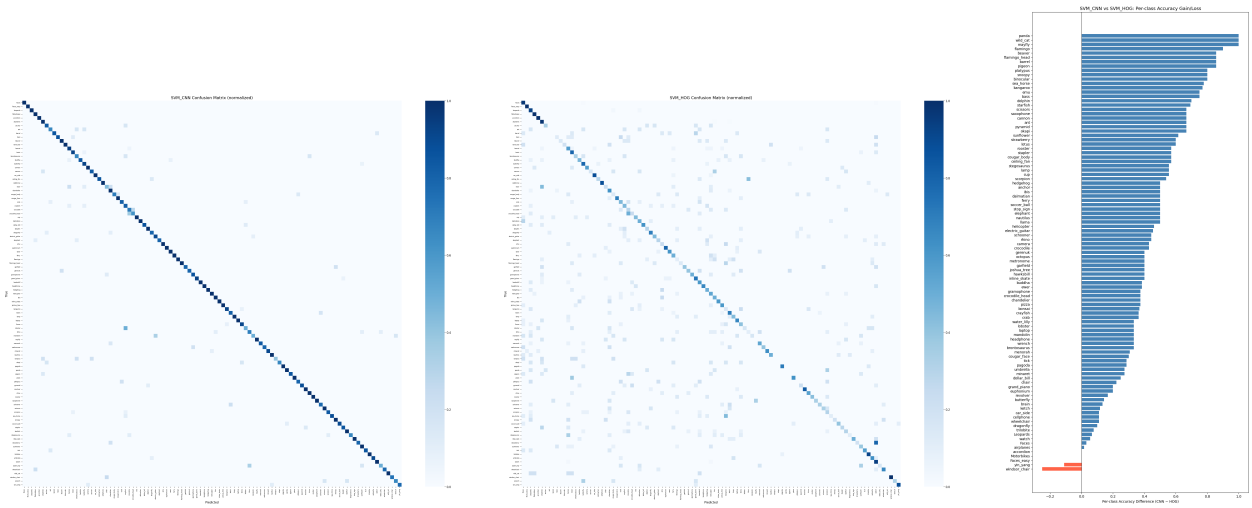
Ablation summary. Performance remains stable between 224×224 and 128×128 but declines sharply at 64×64, confirming sensitivity to reduced spatial resolution. At 224×224, augmentation does not affect overall accuracy, although it yields a marginal improvement in Macro F1, suggesting a minor benefit for minority-class balance without altering top-line performance.

Classical ML: SVM Pipelines

Table 3: Classical ML results comparing HOG and frozen CNN features.

Config	Accuracy	Top-5 Acc	Macro F1	Weighted F1
SVM_HOG	0.6375	0.8187	0.4701	0.6194
SVM_CNN	**0.9309**	**0.9900**	**0.8962**	**0.9300**

Table 3 compares the two classical pipelines. Replacing HOG features with frozen CNN embeddings produces a substantial performance improvement across all metrics. The SVM_CNN configuration achieves 93.09% accuracy and a Macro F1 of 0.8962, whereas SVM_HOG reaches only 63.75% accuracy and 0.4701 Macro F1. Precision and recall trends closely mirror those observed in the F1 scores.



(a) SVM_CNN (b) SVM_HOG (c) Per-class gain

Figure 3: Classical SVM pipeline comparison. (a): confusion matrix using CNN embeddings. (b): confusion matrix using HOG features. (c): per-class accuracy difference (CNN - HOG).

The CNN-based pipeline exhibits strong diagonal concentration, indicating well-separated class representations. In contrast, the HOG-based model shows substantial off-diagonal mass, reflecting weaker feature discriminability.

Figure 3c shows that the CNN-based representation improves accuracy for the vast majority of classes, with only a few categories exhibiting minor degradation.

Representation ablation. Holding the classifier fixed, switching from HOG to CNN features

improves accuracy by approximately 29 percentage points. This result indicates that feature representation quality, rather than classifier capacity, is the primary performance bottleneck in the classical pipeline.

4 Discussion

Deep convolutional models consistently outperformed classical pipelines on Caltech-101. EfficientNet-B0 achieved the strongest overall performance, followed closely by ResNet-18, indicating that modern pretrained architectures provide robust transfer learning baselines for medium-scale image classification tasks.

Effect of Image Resolution: Input resolution is the dominant factor influencing performance among the ablation variables. Reducing resolution from 224×224 to 128×128 causes only moderate degradation, while downsampling to 64×64 results in substantial performance loss across both architectures. This pattern indicates that fine-grained spatial detail is necessary for separating visually similar object categories. The similar degradation magnitude for ResNet-18 and EfficientNet-B0 suggests that the limitation is information-driven rather than architecture-specific.

Effect of Data Augmentation: The impact of augmentation is secondary relative to resolution. For ResNet-18, the non-augmented configuration slightly outperforms the augmented variant, whereas EfficientNet-B0 shows nearly identical performance under both settings. These results suggest that full fine-tuning of pretrained models on moderately sized datasets may reduce the marginal benefit of aggressive augmentation. Overall, augmentation influences optimization behavior but does not materially change final model ranking.

Feature Representation in Classical ML: The representation ablation provides the clearest performance insight. Holding the SVM classifier fixed while replacing HOG descriptors with frozen CNN embeddings yields a dramatic improvement across all metrics. This result demonstrates that feature representation quality, rather than classifier capacity, is the primary bottleneck in the classical pipeline. Learned deep features encode semantic structure that hand-crafted de-

scriptors fail to capture, enabling substantial gains even without end-to-end training.

5 Conclusion

This study compared deep learning and classical machine learning approaches for image classification on Caltech-101 under controlled training conditions. EfficientNet-B0 achieved the best overall performance, with ResNet-18 remaining a strong and competitive baseline. The classical experiments show that replacing HOG descriptors with frozen CNN embeddings produces large performance gains, demonstrating that feature representation quality is the primary bottleneck in traditional pipelines. Resolution ablations confirm that aggressive downsampling substantially harms accuracy, while data augmentation provides only limited additional benefit under full fine-tuning. Overall, results demonstrate the effectiveness of transfer learning and the central role of learned representations in modern computer vision systems.